

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Supplementary Examination Summer– 2024**

**Branch: Electronics and Telecommunication Engineering/ Electronics and
Communication Engineering/ Electronics Engineering**

Course: B. Tech

Sem.:- V

Subject with Subject Code: - Digital Signal Processing BTETC502/ BTEXC502

Marks: 60

Date: 03/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Write the applications of Digital Signal Processing.	CO1	6
B) State and prove any six properties of DTFT.	CO2	6
C) State and prove sampling Theorem with neat diagrams.	CO1	6
Q.2 Solve Any Two of the following.		12
A) What is twiddle factor in discrete Fourier transform? Explain its properties.	CO2	6
B) Perform 4 point Circular convolution of the following two sequences using matrix method. $x_1(n) = \{1, 2, 2\}, 0 \leq n \leq 2$ $x_2(n) = \{1, 2, 3, 4\}, 0 \leq n \leq 3$	CO2	6
C) Determine N-point IDFT of $X(k) = \delta(k), 0 \leq k \leq N-1$ and for $0 \leq k_0 \leq N-1$, determine the DFT of using the previous result. 1) $x_1(n) = e^{j\frac{2\pi}{N}k_0n}$ 2) $x_2(n) = e^{-j\frac{2\pi}{N}k_0n}$ 3) $x_3(n) = \cos\left(\frac{2\pi}{N}k_0n\right)$ 4) $x_4(n) = \sin\left(\frac{2\pi}{N}k_0n\right)$	CO2	6
Q.3 Solve Any Two of the following.		12
A) What is RoC of Z- transform? Explain its properties.	CO3	6
B) Determine the Inverse Z.T. of the following for $ z > 1/3$	CO3	6

$$X(z) = \frac{1 + z^{-1}}{1 - \frac{1}{3}z^{-1}}$$

- C) Determine the Z.T. and sketch the ROC of the following signals: CO3 6
- (a) $x(n) = (0.5)^{-n} u(n) + (2)^{-n} u(n)$
- (b) $x(n) = -(0.5)^{-n} u(-n-1) + (2)^{-n} u(-n-1)$

Q.4 Solve Any Two of the following. 12

- A) Let the lattice coefficients of a three stage FIR lattice structure be $K_1 = 0.1$, $K_2 = 0.2$, $K_3 = 0.3$. Determine the FIR filter coefficients for the Direct form structure. CO5 6
- B) Give comparison of Impulse Invariant Method and Bilinear Transformation Method. CO5 6
- C) Derive expression of $y(n)$ and draw the direct form- I and direct form- II realizations of the transfer function given by CO4 6

$$H(z) = \frac{1 + 2z^{-1} - z^{-2}}{1 + z^{-1} - z^{-2}}$$

Q.5 Solve Any Two of the following. 12

- A) Explain cascade form realization for FIR filters. Derive expression of $y(n)$ and draw the cascade form realizations of the transfer function given by CO4 6

$$H(z) = \left(1 + \frac{1}{5}z^{-1} + z^{-2}\right) \left(1 + z^{-1} + \frac{1}{5}z^{-2}\right)$$

- B) Explain Aliasing effect in downsampling with the help of neat diagrams. Explain the process to prevent this effect. CO5 6
- C) For the transfer function CO4 6

$$H_a(s) = \frac{4s + 7}{s^2 + 5s + 4}$$

Find the corresponding system function $H(z)$ using Impulse Invariant Method. Assume $F_s = 2$ Hz.

*** End ***